REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 10 Advanced | Module 16 Lesson 1 | Reflection of Light

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: θi = θr, measured from the normal. Plane mirror: di = −do; hi = ho; magnification = +1.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 3–8.

P01. The angle of incidence is measured between the incident ray and the:

A. Mirror surface B. Normal
C. Image plane D. Object height

Answer

B. Normal.

P02. For specular reflection, the surface is smooth compared with the:

A. Distance to the Sun B. Observer's height
C. Wavelength of light D. Mirror frame

Answer

C. Wavelength of light.

P03. An image formed by a plane mirror is:

A. Real and inverted B. Real and enlarged
C. Virtual and inverted D. Virtual and upright

Answer

D. Virtual and upright.

P04. Diffuse reflection occurs because:

A. Local surface normals differ across a rough surface B. The law of reflection fails
C. Light loses all wave properties D. Every ray is absorbed

Answer

A. Local surface normals differ across a rough surface.

P05. A plane-mirror image is located:

A. At the mirror surface always B. As far behind the mirror as the object is in front
C. Twice the object distance in front D. At a fixed focal point

Answer

B. As far behind the mirror as the object is in front.

P06. A light ray has incidence angle 20° to the normal at a plane mirror. Find the reflection angle to the normal.

A. 40 ° B. 10 °
C. 20 ° D. -20 °

Answer

C. 20 °. Law of reflection: θr = θi = 20°.

P07. A ray makes 30° with the mirror surface. Find its incidence angle measured from the normal.

A. 120 ° B. 30 °
C. -60 ° D. 60 °

Answer

D. 60 °. Normal is perpendicular to surface, so θi = 90° − 30° = 60°.

P08. A ray strikes a plane mirror at incidence angle 20°. Find the smaller angle between the incident and reflected ray arms at the surface.

A. 40 ° B. 80 °
C. 20 ° D. -40 °

Answer

A. 40 °. The two ray arms make equal angles with the normal. Included angle = 2θi = 2 × 20° = 40°.

P09. An object is 24 cm in front of a plane mirror. Find the distance between the object and its image.

A. 96 cm B. 48 cm
C. 24 cm D. -48 cm

Answer

B. 48 cm. The image is 24 cm behind the mirror. Object–image separation = 24 + 24 = 48 cm.

P10. An object moves 6 cm closer to a stationary plane mirror. By how much does its separation from the image decrease?

A. 24 cm B. 6 cm
C. 12 cm D. -12 cm

Answer

C. 12 cm. Object and image each move 6 cm toward the mirror; separation decreases by 2 × 6 = 12 cm.

P11. State the law of reflection.

Answer

The incident ray, reflected ray and normal lie in one plane, and the incidence angle equals the reflection angle.

P12. Distinguish specular reflection from diffuse reflection.

Answer

Specular reflection from a smooth surface preserves an organised ray pattern; diffuse reflection from a rough surface scatters rays in different directions.

P13. Why is a plane-mirror image virtual?

Answer

Reflected rays do not actually meet behind the mirror; their backward extensions appear to meet there.

P14. State the size and orientation of a plane-mirror image.

Answer

It has the same height as the object and is upright, with lateral reversal relative to the observer's usual perspective.

P15. Why can a rough page be seen from many directions?

Answer

It diffusely reflects incident light into a range of directions.

P16. A light ray has incidence angle 25° to the normal at a plane mirror. Find the reflection angle to the normal.

Answer

Law of reflection: $\theta r = \theta i = 25°$.

P17. A light ray has incidence angle 30° to the normal at a plane mirror. Find the reflection angle to the normal.

Answer

Law of reflection: $\theta r = \theta i = 30°$.

P18. A ray makes 35° with the mirror surface. Find its incidence angle measured from the normal.

Answer

Normal is perpendicular to surface, so $\theta i = 90° - 35° = 55°$.

P19. A ray makes 40° with the mirror surface. Find its incidence angle measured from the normal.

Answer

Normal is perpendicular to surface, so $\theta i = 90° - 40° = 50°$.

P20. A ray strikes a plane mirror at incidence angle 25°. Find the smaller angle between the incident and reflected ray arms at the surface.

Answer

The two ray arms make equal angles with the normal. Included angle = $2\theta i = 2 \times 25° = 50°$.

P21. A ray strikes a plane mirror at incidence angle 30°. Find the smaller angle between the incident and reflected ray arms at the surface.

Answer

The two ray arms make equal angles with the normal. Included angle = $2\theta i = 2 \times 30° = 60°$.

P22. An object is 36 cm in front of a plane mirror. Find the distance between the object and its image.

Answer

The image is 36 cm behind the mirror. Object–image separation = 36 + 36 = 72 cm.

P23. An object is 48 cm in front of a plane mirror. Find the distance between the object and its image.

Answer

The image is 48 cm behind the mirror. Object–image separation = 48 + 48 = 96 cm.

P24. An object moves 9 cm closer to a stationary plane mirror. By how much does its separation from the image decrease?

Answer

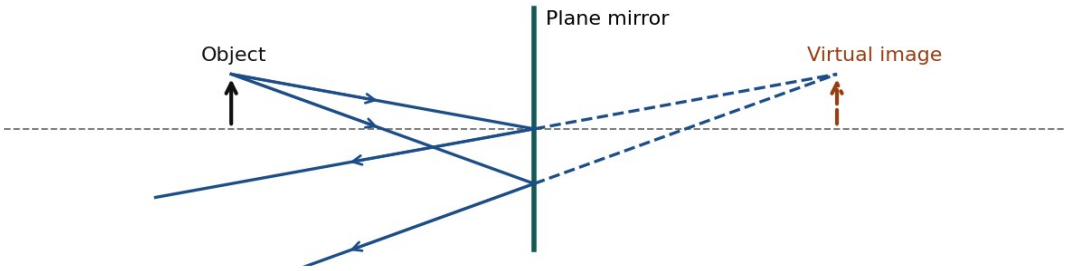
Object and image each move 9 cm toward the mirror; separation decreases by 2 × 9 = 18 cm.

P25. An object moves 12 cm closer to a stationary plane mirror. By how much does its separation from the image decrease?

Answer

Object and image each move 12 cm toward the mirror; separation decreases by 2 × 12 = 24 cm.

P26. Draw a plane mirror, an object point and two rays to locate its image. Label the normal for one ray.



Answer

Reflect both rays using equal incidence and reflection angles. Extend the reflected rays backward with dashed lines; their extensions meet behind the mirror at equal perpendicular object and image distances.

P27. A student measures 35° between a ray and the mirror surface and calls it the incidence angle. Correct the value.

Answer

The incidence angle is measured from the normal, so it is 90° − 35° = 55°. The reflection angle is also 55°.

P28. Does diffuse reflection violate the law of reflection? Explain.

Answer

No. Each ray obeys the law relative to the local normal. A rough surface has different local normals, so the outgoing rays are spread out.

P29. Why does moving a plane-mirror object closer not make its image physically larger?

Answer

A plane mirror has magnification +1, so image height equals object height. Its apparent angular size to an observer can still change with viewing geometry.

P30. Design a simple experiment to test the law of reflection.

Answer

Place a plane mirror on paper, draw a normal at the incidence point, trace narrow incident and reflected beams and measure their angles from the normal for several trials.